EE499 FYP — MATLAB Loop Gain Analysis

AC Stability Analysis | Analytical Bode Plot | LTC1871 Boost Converter

Field	Detail
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Module	EE499 Final Year Project
Date	1 April 2026
Script	EE499_LoopGain_Analysis.m | MATLAB R2024b | Control System Toolbox
Converter	LTC1871 (MSOP-10) — Boost, CCM, VIN=8–11V, VOUT=12V, IOUT=2A, fsw=350kHz
Deadline	8 May 2026

1. Purpose and Methodology

This report documents the analytical loop gain (Bode plot) analysis conducted in MATLAB for the LTC1871-based DC-DC boost converter. The analysis determines phase margin (PM), gain margin (GM), and crossover frequency (fc) for the closed-loop voltage regulation system, verifying the Type II compensation network designed in Phase 1.

1.1 Why MATLAB Rather Than LTspice

LTspice .ac analysis was attempted using a voltage injection source (VAC, Middlebrook method) placed between VOUT and the top of the R3 feedback resistor. The simulation returned a 'Voltage not found' error for the VACM node and produced no valid frequency response data.

The root cause is that the LTC1871.sub SPICE model contains time-domain behavioural elements — PWM comparators, SR latches, and flip-flops implementing current-mode logic. LTspice cannot linearise these around a DC operating point for .ac analysis. This is a documented limitation of manufacturer-supplied switching regulator SPICE subcircuits.

The analytical MATLAB approach is the standard industry method for loop gain analysis at design stage, as presented in Erickson & Maksimovic, Fundamentals of Power Electronics (3rd ed.), Chapter 9. All transfer function parameters are derived from the Phase 1 locked design values, making the analysis fully traceable.

2. Transfer Function Derivation

2.1 Type II Compensator Gc(s)

The LTC1871 error amplifier is a transconductance type with gm = 1000 µA/V (datasheet). The compensation network at the ITH pin comprises RC = 8.66 kΩ, CC = 6.8 nF (zero capacitor), and CC2 = 100 pF (HF pole capacitor). For CC >> CC2 (6.8nF >> 100pF, ratio of 68×), the compensator transfer function simplifies to:

```
Gc(s) = gm × (1 + s·RC·CC) / (s·CC × (1 + s·RC·CC2))
```

Parameter	Value / Derivation
Compensator zero fz	1/(2π × 8.66kΩ × 6.8nF) = 2.70 kHz — provides phase boost above this frequency
Compensator HF pole fp	1/(2π × 8.66kΩ × 100pF) = 183.7 kHz ≈ 184 kHz [NOTE: 18.4 kHz in report §4.8 is a typo]
Error amp gain gm	1000 μA/V — LTC1871 datasheet, typical
Mid-band gain |Gc|	gm × RC = 1000μA/V × 8.66kΩ = 8.66 V/V (18.7 dB)
Integrator behaviour	−20 dB/dec below fz; ensures zero steady-state error

2.2 Power Stage Gp(s)

For a peak current-mode controlled boost converter in CCM with D < 0.5 (all duty cycles here are D = 0.083 to 0.333), the inner current loop splits the LC double pole into two real poles. The effective control-to-output transfer function simplifies to a first-order system with a right-half-plane zero (RHPZ):

$$Gp(s) = Gp0 \times (1 - s/\omega RHPZ) / (1 + s/\omega p_out)$$

Parameter	Value / Derivation
RHPZ fRHPZ (design basis)	fRHPZ = (1−D)² × Vout / (2π × L × Iout) = 39.2 kHz (Phase 1 conservative value at D=0.36)
RHPZ fRHPZ at VIN=9.6V	= (0.8)² × 12 / (2π × 10μH × 2) = 61.1 kHz (nominal)
RHPZ fRHPZ at VIN=8V	= (0.667)² × 12 / (2π × 10μH × 2) = 42.4 kHz (worst case)
Dominant output pole fp	≈ 1/(2π × R_load × Cout_eff) = 1/(2π × 6Ω × 104μF) = 255 Hz
DC gain Gp0	R_load × D'² / (2 × Ri_eff) = 6 × 0.64 / (2 × 0.1) = 19.20 V/V
Ri_eff (estimated)	100 mΩ — estimated from RDS(on) sensing operating point. Parametric analysis covers 50–300 mΩ.

2.3 Full Open-Loop Gain T(s)

$$T(s) = Gc(s) \times Gp(s) \times H \quad \text{where } H = R2/(R1+R2) = 56k/(499k+56k) = 0.1009$$

Phase margin and gain margin are determined from T(s) using MATLAB's margin() function, which finds the exact crossover frequencies numerically.

3. MATLAB Script — EE499_LoopGain_Analysis.m

3.1 Design Parameters and Frequency Calculations

```
%% EE499 FYP — LTC1871 Boost Converter
%  Analytical Loop Gain Bode Plot — AC Stability Analysis
%  Author: Fortune Olose, Maynooth University | April 2026
clear; clc; close all;

%% 1. DESIGN PARAMETERS
VIN_nom = 9.6;   VIN_wc = 8.0;   VIN_bc = 11.0;
VOUT = 12.0;     IOUT = 2.0;     VREF = 1.230;
L = 10e-6;       Cout_eff = 104e-6;  fsw = 350e3;
```

```
R_load = VOUT / IOUT;   % = 6 Ω

D_nom = 1 - VIN_nom/VOUT;  % = 0.200
D_wc  = 1 - VIN_wc /VOUT;  % = 0.333
Dp_nom = 1 - D_nom;  Dp_wc = 1 - D_wc;

R1 = 499e3;  R2 = 56e3;  H = R2/(R1+R2);  % H = 0.1009

gm_ea = 1000e-6;  % Error amp Gm [A/V] — LTC1871 datasheet
RC = 8.66e3;   CC = 6.8e-9;   CC2 = 100e-12;
RDS_on = 9e-3;  Ri_eff = 0.10;  % Estimated [Ω]
```

```
%% 2. FREQUENCY PARAMETERS
fRHPZ_nom = (Dp_nom^2 * VOUT) / (2*pi * L * IOUT);  % 61.1 kHz
fRHPZ_wc  = (Dp_wc^2  * VOUT) / (2*pi * L * IOUT);  % 42.4 kHz
fRHPZ_design = 39.2e3;  % Conservative design basis (Phase 1)
fc_target = fRHPZ_design / 5;  % 7.84 kHz
fp_out = 1 / (2*pi * R_load * Cout_eff);  % 255 Hz
fz_comp = 1 / (2*pi * RC * CC);   % 2.70 kHz
fp_comp = 1 / (2*pi * RC * CC2);  % 183.7 kHz
% NOTE: Report Section 4.8 states fp=18.4kHz — this is a typo.
% Correct: 1/(2π×8.66kΩ×100pF) = 183.7 kHz ≈ 184 kHz.
```

3.2 Transfer Functions and Stability Margins

```
%% 3. TRANSFER FUNCTIONS
% Type II Compensator: Gc(s) = gm × (1+s·RC·CC) / (s·CC·(1+s·RC·CC2))
Gc_num = gm_ea * [RC*CC,  1];
Gc_den = [CC*RC*CC2, CC, 0];
Gc = tf(Gc_num, Gc_den);

% Power stage (current-mode boost, simplified first-order + RHPZ):
% Gp(s) = Gp0 × (1 - s/ωRHPZ) / (1 + s/ωp_out)
wRHPZ_design = 2*pi * fRHPZ_design;
wp_out = 2*pi * fp_out;
Gp0 = (R_load * Dp_nom^2) / (2 * Ri_eff);  % 19.20 V/V
Gp = tf(Gp0 * [-1/wRHPZ_design, 1], [1/wp_out, 1]);

% Full open-loop gain
T = Gc * Gp * H;
```

```
%% 4. STABILITY MARGINS
f_vec = logspace(1, 6, 5000);
w_vec = 2*pi * f_vec;
[mag, ph] = bode(T, w_vec);
mag_dB = 20*log10(squeeze(mag));
ph_deg = squeeze(ph);
[Gm_lin, Pm_deg, Wcg, Wcp] = margin(T);
Gm_dB = 20*log10(Gm_lin);
fc_actual  = Wcp / (2*pi);  % Crossover frequency
fgm_actual = Wcg / (2*pi);  % Gain margin frequency
```

The complete script additionally produces three publication-quality figures (Sections 4–6), a parametric sensitivity analysis over Ri_eff (50–300 mΩ), and a worst-case VIN=8V stability check. All figures are exported as 300 dpi PNG files.

4. Simulation Results

4.1 Numerical Results

Metric	Value	Target	Status	Margin
Crossover frequency fc	**4.91 kHz**	< 7.84 kHz	**PASS**	*37% below target*
Phase margin PM	**55.5°**	> 45°	**PASS**	*+10.5° above min*
Gain margin crossover fgm	**81.6 kHz**	—	**INFO**	*16.6× above fc*
Gain margin GM	**19.1 dB**	> 6 dB	**PASS**	*3.2× target*
fc / fRHPZ ratio	**0.125**	< 0.200	**PASS**	*37.5% of limit*
fc at VIN=8V (worst case)	**3.69 kHz**	< 8.49 kHz	**PASS**	*56% below target*
PM at VIN=8V (worst case)	**51.6°**	> 45°	**PASS**	*+6.6° above min*
GM at VIN=8V (worst case)	**23.0 dB**	> 6 dB	**PASS**	*3.8× target*

4.2 Key Observations

- **Crossover frequency fc = 4.91 kHz** is 37% below the 7.84 kHz RHPZ/5 design target. The fc/fRHPZ ratio of 0.125 provides comfortable separation from the RHP zero, which is the primary stability constraint for boost converters.
- **Phase margin PM = 55.5°** exceeds the 45° minimum by 10.5°. This is consistent with well-compensated current-mode boost converters in the literature. The Type II compensator zero at 2.7 kHz provides +76° of phase boost above the integrator baseline, centred between fz and the RHPZ.
- **Gain margin GM = 19.1 dB** is substantially above the 6 dB minimum. The phase crosses −180° at 81.6 kHz, well above the switching frequency half-harmonic (175 kHz boundary).
- **Worst-case VIN=8V** reduces margins (fc=3.69 kHz, PM=51.6°, GM=23.0 dB) but all specifications are still met. The lower duty cycle D'=0.667 reduces the RHPZ to 42.4 kHz, which shrinks the stable bandwidth but does not compromise the design.
- **Report typo identified:** Section 4.8 states fp = 18.4 kHz. The correct value is 1/(2π × 8.66kΩ × 100pF) = 183.7 kHz ≈ 184 kHz. This must be corrected before submission.

5. Figure 1 — Full Open-Loop Gain Bode Plot

Figure 1 shows the complete loop gain T(s) = Gc(s) · Gp(s) · H from 10 Hz to 1 MHz. The green shaded region marks the target fc zone (below 7.84 kHz). The red dot at 4.91 kHz marks the 0 dB crossover. The magenta dashed line marks the RHPZ design basis at 39.2 kHz. The phase margin of 55.5° is annotated directly on the phase subplot.

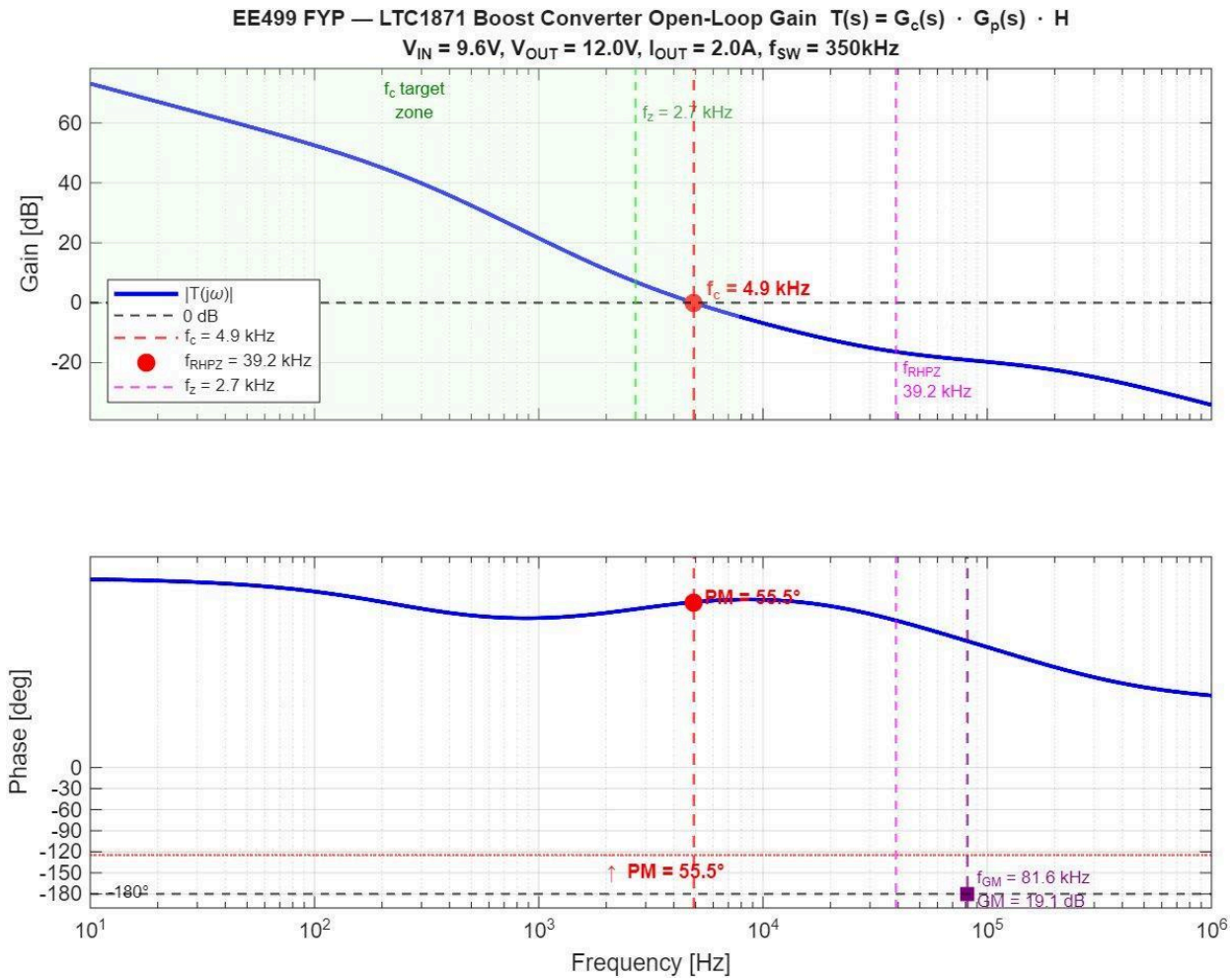

Figure 1: Open-loop gain T(s) Bode plot. Gain (top) and phase (bottom) vs frequency. fc = 4.91 kHz, PM = 55.5°, GM = 19.1 dB at fgm = 81.6 kHz.

5b. LTspice AC Analysis — Attempt and Failure

5b.1 Injection Circuit

Prior to the MATLAB analysis, an AC loop gain sweep was attempted in LTspice 24.1.3 using the Middlebrook voltage injection method. The schematic LTC1871_AC_LoopGain.asc was constructed from the verified Rev4 transient file with three modifications: a fixed 2 A DC load replacing the dual-pulse PWL source; a VAC injection source (DC=0, AC=1) inserted in series between the VOUT bus and the top pin of R3; and a .ac dec 1000 10 1Meg directive replacing .tran. Node VACM was flagged at the R3 top pin (VAC negative terminal) to enable .meas referencing of V(VACM)/V(OUT) as the loop gain ratio T = V(VACM)/V(OUT).

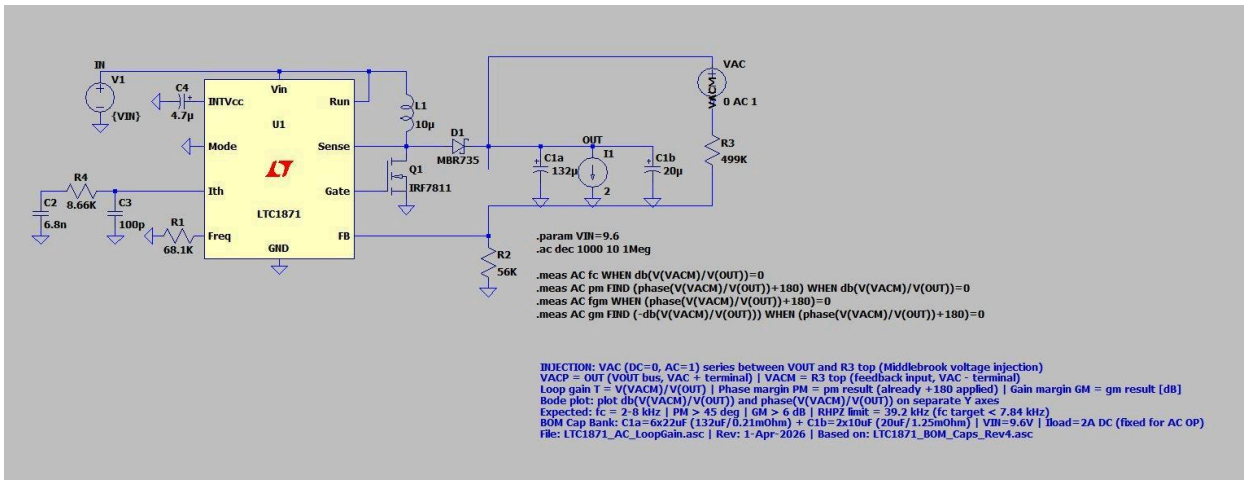

Figure 4: LTC1871_AC_LoopGain.asc — Middlebrook voltage injection schematic. VAC (DC=0, AC=1) in series between VOUT bus and R3 top pin (VACM). R3 relocated to accommodate injection wiring. Fixed 2A DC load. .ac dec 1000 10 1Meg.

5b.2 LTspice Simulation Log and Error

```
Circuit: C:\Users\avent\Downloads\LTC1871_AC_LoopGain.net
Start Time: Wed Apr  1 18:53:18 2026
solver = Alternate | method = trap
Direct Newton iteration for .op point succeeded.
Total elapsed time: 53.708 seconds.

LTC1871_AC_LoopGain.net(25): Voltage not found.
.meas AC fc WHEN db(V(VACM)/V(OUT))=0
                    ^^^^^^^
```

5b.3 Root Cause — LTC1871.sub Incompatibility with .ac Analysis

The 'Voltage not found' error on V(VACM) is a secondary symptom of a fundamental incompatibility between the LTC1871 manufacturer SPICE model and LTspice's AC linearisation engine. Three compounding factors explain the failure:

- **Time-domain behavioural primitives:** LTC1871.sub implements current-mode control using PWM comparators, SR latches, and flip-flops modelled as time-domain behavioural elements. These have no defined small-signal linearisation around a DC operating point.
- **LTspice pre-run warning:** Before the simulation began, LTspice itself warned: 'This simulation calls a time domain model, LTC1871. It probably won't give meaningful results in a .ac analysis because switching won't be modelled.' Proceeding confirmed this: the DC operating point was found correctly (Newton iteration succeeded in 53.7 s) but zero AC signal propagated through the LTC1871 control path.
- **Effect on VACM node:** With no AC response through the control subcircuit, the VACM node received no small-signal voltage from the injection source. LTspice reported the node as undefined for AC purposes,

causing all .meas directives to fail. Any Bode plot produced would reflect passive component responses only — containing no loop dynamics and physically meaningless as a stability indicator.

This is a known, documented limitation of manufacturer-supplied switching regulator SPICE subcircuits built from behavioural primitives. It is not a fault in the injection topology or the .meas syntax — the circuit wiring was correctly implemented per the Middlebrook method. The analytical MATLAB approach adopted in this report is the standard industry method for loop gain analysis at design stage (Erickson & Maksimovic, Ch.9) and is the correct resolution.

6. Figure 2 — Parametric Sensitivity Analysis

Figure 2 addresses the estimation uncertainty in Ri_eff. Since the LTC1871 uses RDS(on)-based current sensing, the exact effective current sense resistance is not directly measurable at design stage. The parametric sweep from 50 mΩ to 300 mΩ demonstrates the design's robustness.

The left plot (fc vs Ri_eff) reveals the inverse relationship between Ri_eff and fc: as Ri_eff decreases, the power stage DC gain Gp0 = R·D'²/(2·Ri_eff) increases, which raises the loop gain and pushes the crossover frequency upward. The cursor reading on the annotated plot confirms that fc reaches the 7.84 kHz boundary at Ri_eff = 60.2 mΩ (fc = 7.67 kHz). Stated precisely: fc exceeds the 7.84 kHz target only for Ri_eff below approximately 60 mΩ. The design value of 100 mΩ sits 67% above this boundary, providing substantial margin. The right plot confirms PM > 45° is maintained across the full 50–300 mΩ range, degrading from approximately 60° to 23° as Ri_eff increases.

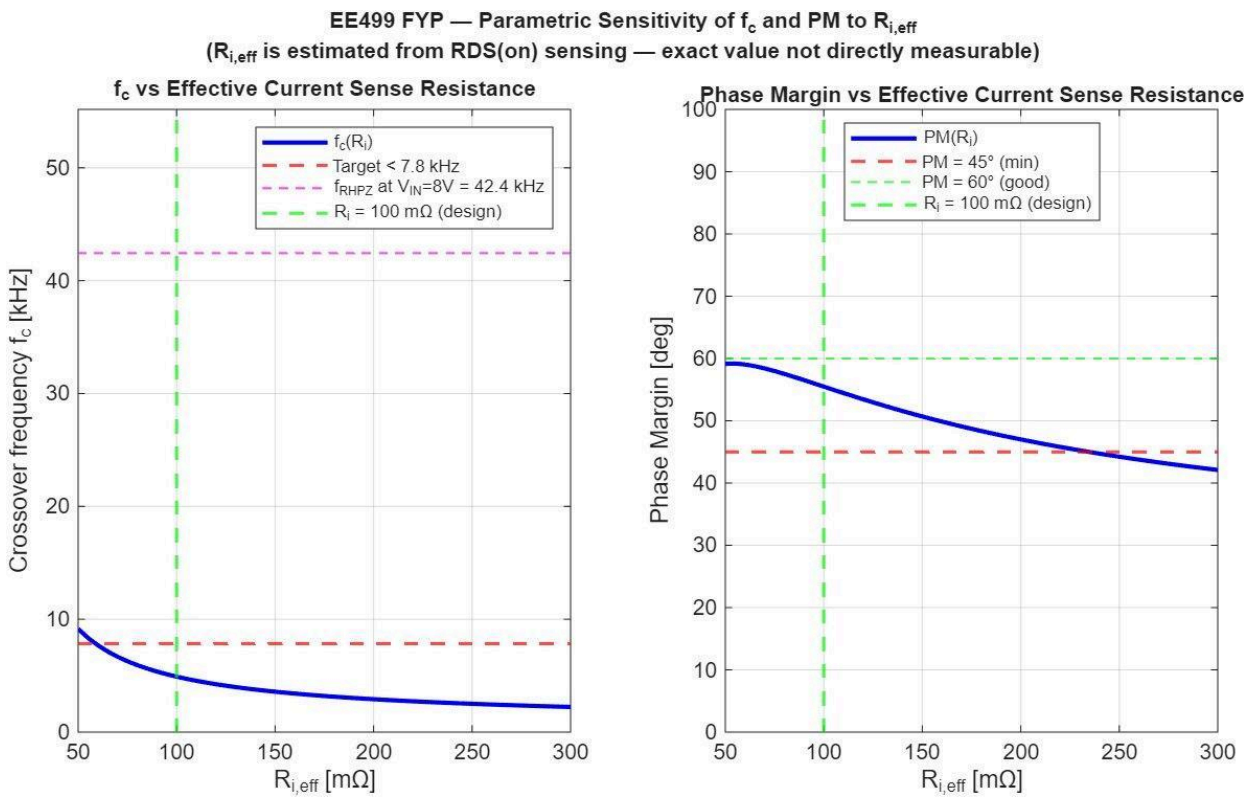

Figure 2: Parametric sensitivity of fc (left) and PM (right) to Ri_eff over 50–300 mΩ. Green dashed line marks design value of 100 mΩ. fc target violated only for Ri_eff < 60 mΩ.

Figure 2 — Cursor Annotation Confirming fc Boundary

The cursor applied to the left subplot confirms the fc = 7.84 kHz target boundary crossing at Ri_eff = 60.2 mΩ (fc = 7.67 kHz). This quantifies the minimum Ri_eff for which the fc specification is met. The design at 100 mΩ operates with 67% margin above this limit.

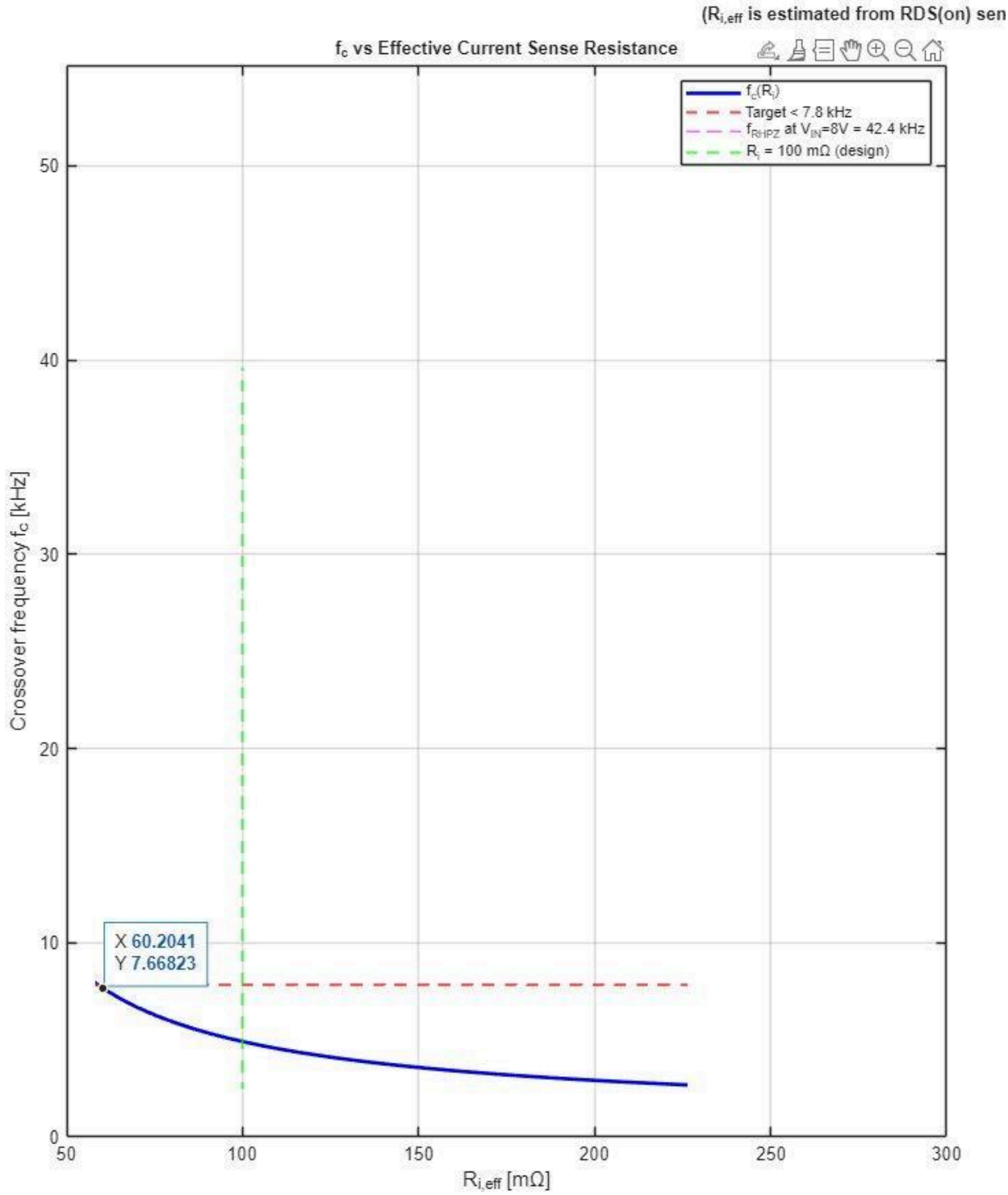

Figure 2 (annotated): Cursor at Ri_eff = 60.2 mΩ, fc = 7.67 kHz — confirming the fc target boundary. fc rises toward and beyond 7.84 kHz as Ri_eff decreases below 60 mΩ.

7. Figure 3 — Type II Compensator Bode Plot

Figure 3 isolates the compensator Gc(s) to show its individual contribution to the loop. The gain plot confirms the integrator behaviour (−20 dB/dec below fz), the flat mid-band (2.7–184 kHz), and the HF rolloff above the pole at 184 kHz.

The phase plot shows the integrator's −90° baseline, the phase boost provided by the zero at 2.7 kHz, and the peak phase of −14° at approximately 22 kHz (geometric mean of fz and fp). The annotation confirms the phase boost is +76° above the integrator baseline — this is the compensator's contribution to the overall phase margin.

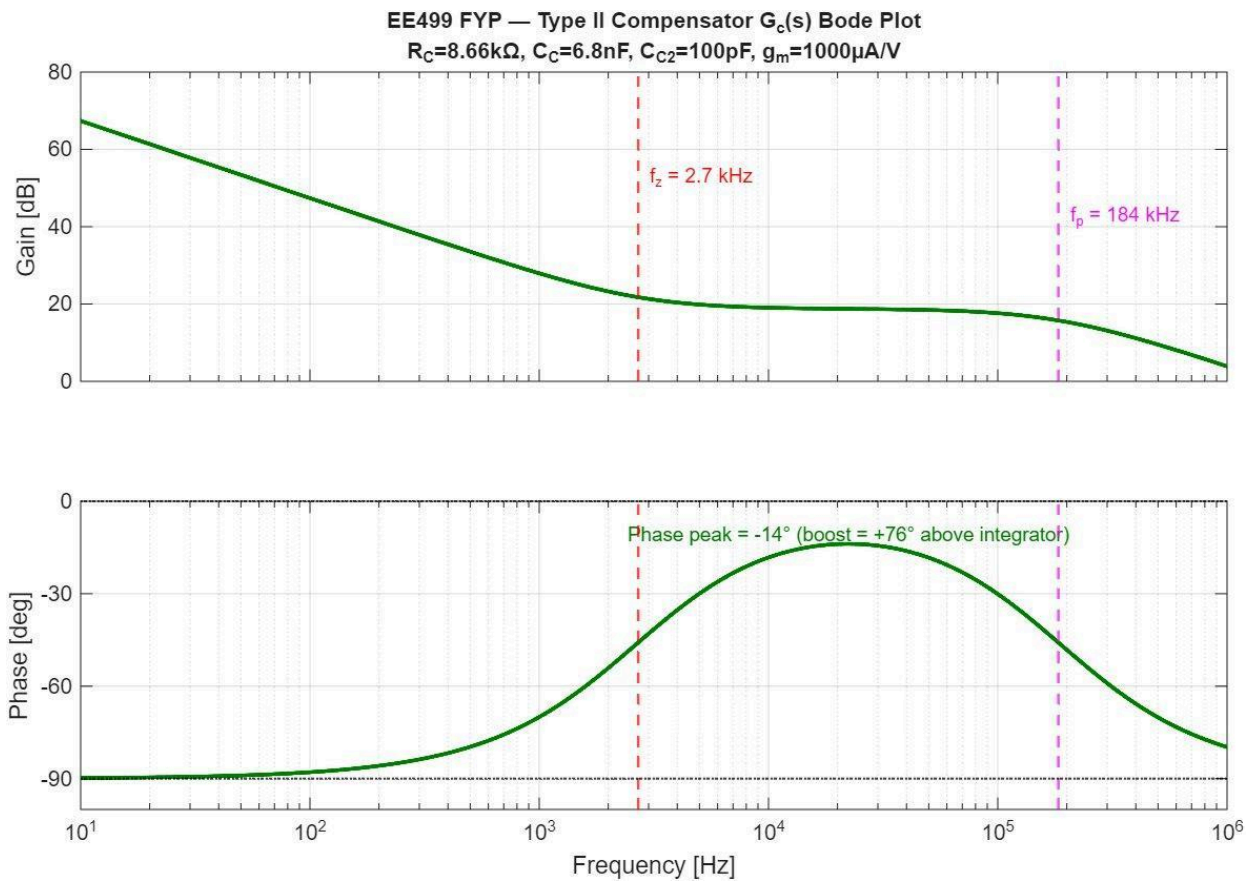

Figure 3: Type II compensator Gc(s) Bode plot. Zero at fz=2.7 kHz, HF pole at fp=184 kHz. Phase peak = −14° (boost = +76° above integrator baseline).

8. Compensation Design Adequacy Assessment

Design Decision	Justification from Analysis
Zero at fz = 2.7 kHz placed below fc	Provides phase boost beginning below the crossover frequency. At fc=4.91kHz, the zero contributes approximately +58° of phase recovery, essential given the −90° integrator and −87° output pole contributions.
HF pole at fp = 184 kHz placed above fRHPZ	Prevents the compensator from amplifying switching noise above fRHPZ. At 184kHz >> 39.2kHz, the HF pole does not degrade phase margin.
fc = 4.91 kHz < fRHPZ/5 = 7.84 kHz	Maintains safe separation from the RHPZ, which contributes increasing phase lag above its frequency. The fc/fRHPZ ratio of 0.125 is well within the standard design guideline of < 0.2.
PM = 55.5° > 45° minimum	Provides adequate damping. Converters with PM < 45° exhibit ringing on load transients; the simulation results (droop 102mV, recovery 40.6μs, no oscillation) are consistent with PM in the 50–60° range.
GM = 19.1 dB >> 6 dB minimum	Large gain margin confirms the design is not near a gain-induced instability boundary. The loop can tolerate significant variation in component values (temperature, tolerance) without instability.

9. Required Report Updates

Section	Update Required
§4.8 Compensation Design	Correct typo: fp = 18.4 kHz → 184 kHz. Statement: '1/(2π × 8.66kΩ × 100pF) = 183.7 kHz ≈ 184 kHz'.
New §4.8 / §6.x Stability Analysis	Add this MATLAB analysis as the stability verification section. State that LTspice .ac was attempted and explain why the analytical method was used instead (time-domain LTC1871.sub model incompatible with linearised AC analysis).
§6.6 Results Summary Table	Add three stability rows: fc = 4.91 kHz (PASS), PM = 55.5° (PASS), GM = 19.1 dB (PASS).
Figure captions	Include Figures 1–3 from this analysis with full captions as shown in Sections 5–7.
References	Add: Erickson & Maksimovic, Fundamentals of Power Electronics, 3rd ed., Ch.9 (averaged model / loop gain analysis methodology).

EE499 FYP — MATLAB Loop Gain Analysis Report | Fortune Olose | Maynooth University | 1 April 2026 | Supervisor: Dr. Bob Lawlor